allow us to pack the crates with the largest possible total of calories. The problem statement now becomes:

“Given a set of items, choose any subset of items **or fraction** of items from it, such that their total weight won’t be over 1000 kgs, and in such a way that the total amount of calories is maximized”.

And yes, it is worth to go through the overhead of re-packing everything, because we get a nice improvement.

Specifically, if we can take any fraction of the original weights for each product, we can simply pack products starting from the one with the highest ratio of calories per kg (peanut butter, in this example), and when we get to one box that would not fit in the remaining available space, we take a fraction of it to fill the gap, and repack it. So, in the end we wouldn’t even have to repack all the goods, but just one.

The best solution would then be: peanut butter, rice, wheat flour, strawberry jam, and 230 kilograms of beans, for a total of **3,342,800 calories**.

1.4.5 Happy Ending

So, in our story, the future Mars settlers will have greater chances of survival, and won’t go into depression because of a diet only comprising potatoes with a peanut butter and strawberry dip.

From a computational point of view, we moved from incorrect algorithms (the greedy solutions taking the highest